Analysis 11: Log-Power Weights, Crossing Scale, and Concentration for Weighted Rolling Validation

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1 Setup

$$Y_i = \theta + \xi_i, \qquad \xi_i \overset{\text{iid}}{\sim} N(0,1), \qquad 0 < |\theta| < 1.$$

Here Y_i is the observation at time i, θ is the unknown mean, and ξ_i are independent standard Gaussian errors. For current training-sample size $n \in \{1, 2, \ldots\}$, the training sample mean is

$$\bar{Y}_n = \frac{1}{n}\sum_{j=1}^{n} Y_j.$$

Model 0 predicts $\bar{Y}_n$; Model 1 predicts zero. Both use squared loss.

Write $R_k(n)$ for Model k's expected squared prediction error on an independent new observation, and $\Delta_R(n) = R_0(n) - R_1(n)$ for the population-risk difference. Then

$$R_0(n) = 1 + \frac{1}{n}, \qquad R_1(n) = 1 + \theta^2, \qquad \Delta_R(n) = \frac{1}{n} - \theta^2.$$

Define n_{pop} as the first training size with $\Delta_R(n) < 0$:

$$n_{\text{pop}} = \lfloor \theta^{-2} \rfloor + 1 \sim \theta^{-2} \qquad (|\theta| \to 0).$$

Proof. See Appendix A.1.

2 Weighted rolling validation

Rolling validation (RV) predicts Y_i from the first $i-1$ observations. Let $w_i > 0$ be the deterministic weight assigned to validation time i, whose training size is $i-1$; the weight sequence is independent of the data and θ. Define the expected weighted-RV loss difference (Model 0 minus Model 1) by

$$\Delta_{\text{RV},w}(n) = \sum_{i=2}^{n+1} w_i \left(\frac{1}{i-1} - \theta^2 \right).$$

The quantity r_n is the weighted average of the reciprocal training sizes:

$$r_n = \frac{\sum_{i=2}^{n+1} w_i/(i-1)}{\sum_{i=2}^{n+1} w_i}, \qquad \frac{\Delta_{\text{RV},w}(n)}{\sum_{i=2}^{n+1} w_i} = r_n - \theta^2.$$

Define $n_{\text{RV},w}$ as the first training size with $r_n < \theta^2$, provided such an integer exists. Positivity of the weights gives

$$r_n \geq \frac{1}{n}, \qquad n_{\text{RV},w} \geq n_{\text{pop}}.$$

The crossing inequality holds whenever the weighted crossing exists. Also r_n strictly decreases, so there is a unique first strict crossing and the inequality persists afterward. **Proof.** See Appendix A.2.

3 Log-power weights and crossing scale

Take

$$w_i = [\log(i+1)]^p, \qquad p > 0 \text{ fixed.}$$

The exponent p controls the increase in weight given to later validation times. As $n \to \infty$, integral comparison gives

$$\sum_{i=2}^{n+1} w_i \sim n(\log n)^p, \qquad \sum_{i=2}^{n+1} \frac{w_i}{i-1} \sim \frac{(\log n)^{p+1}}{p+1}.$$

Therefore,

$$r_n \sim \frac{\log n}{(p+1)n}.$$

Proof. See Appendix B.1.

Theorem 3.1 (First strict log-power crossing)**.** For fixed $p > 0$, the expected weighted-RV crossing exists for every $0 < |\theta| < 1$, and as $|\theta| \to 0$,

$$n_{\mathrm{RV},w} \sim \frac{\theta^{-2}\log(\theta^{-2})}{p+1} = \frac{2\theta^{-2}}{p+1}\log(|\theta|^{-1}),$$

$$\frac{n_{\mathrm{RV},w}}{n_{\mathrm{pop}}} \sim \frac{\log(\theta^{-2})}{p+1} = \frac{2}{p+1}\log(|\theta|^{-1}).$$

Log-power weighting delays the expected RV crossing by a logarithmic factor relative to the population crossing. **Proof.** See Appendix B.2.

4 Concentration on the log-weight crossing scale

Throughout this section, $p > 0$ is fixed and $|\theta| \to 0$. Suppose

$$\frac{n}{n_{\mathrm{RV},w}} \to c, \qquad 0 < c < \infty, \tag{1}$$

where n is the current training size, $n_{\mathrm{RV},w}$ is the expected weighted-RV crossing size, and c is fixed. Using

$$n_{\mathrm{RV},w} \sim \frac{\theta^{-2}\log(\theta^{-2})}{p+1},$$

we obtain

$$n \sim \frac{c}{p+1}\theta^{-2}\log(\theta^{-2}).$$

Multiplying by θ^2 gives

$$n\theta^2 \sim \frac{c}{p+1}\log(\theta^{-2}).$$

Taking logarithms instead gives

$$\log n = \log(\theta^{-2}) + \log\log(\theta^{-2}) + O(1) \sim \log(\theta^{-2}).$$

The $\log\log(\theta^{-2})$ term is lower order. Here $c < 1$ is before the expected crossing, $c = 1$ is on the crossing scale, and $c > 1$ is after the expected crossing.

4.1 Realized weighted-RV score

Define the realized weighted-RV loss difference

$$\Xi_{n,w} = \sum_{i=2}^{n+1} w_i \left\{ (Y_i - \bar{Y}_{i-1})^2 - Y_i^2 \right\}.$$

Its sign determines the selected model: a negative value favors Model 0, a positive value favors Model 1, and zero is a tie.

4.2 Mean on the log-weight scale

The exact mean identity is

$$\mathbb{E}\Xi_{n,w} = \Delta_{\mathrm{RV},w}(n) = \left(\sum_{i=2}^{n+1} w_i \right) (r_n - \theta^2).$$

For log-power weights,

$$\sum_{i=2}^{n+1} w_i \sim n(\log n)^p, \qquad r_n \sim \frac{\log n}{(p+1)n}.$$

Thus the two contributions satisfy

$$\left(\sum_{i=2}^{n+1} w_i \right) r_n \sim \frac{(\log n)^{p+1}}{p+1},$$

$$\theta^2 \sum_{i=2}^{n+1} w_i \sim n\theta^2 (\log n)^p.$$

Using the scaling relations established above, for fixed $c \neq 1$,

$$\mathbb{E}\Xi_{n,w} \sim \frac{1-c}{p+1} [\log(\theta^{-2})]^{p+1}.$$

The mean is eventually positive for $c < 1$ and negative for $c > 1$; at $c = 1$ the leading terms cancel.
Proof. See Appendix C.1.

4.3 Variance

Start from the centered realized score

$$\Xi_{n,w} - \mathbb{E}\Xi_{n,w}.$$

Expanding the squared-loss difference gives two uncorrelated random components: a quadratic Gaussian part and the linear Gaussian term

$$-2\theta \sum_{i=2}^{n+1} w_i \xi_i.$$

Because these components are uncorrelated,

$$\mathrm{Var}(\Xi_{n,w}) = \mathrm{Var}(\text{quadratic part}) + \mathrm{Var}\left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i \right).$$

The quadratic variance satisfies

$$\mathrm{Var}(\text{quadratic part}) \sim 2(\log n)^{2p}.$$

For the linear component, the square-weight asymptotic

$$\sum_{i=2}^{n+1} w_i^2 \sim n(\log n)^{2p}$$

gives

$$\mathrm{Var}\left(-2\theta \sum_{i=2}^{n+1} w_i \xi_i\right) = 4\theta^2 \sum_{i=2}^{n+1} w_i^2 \sim 4n\theta^2 (\log n)^{2p}.$$

Using the scaling relations established above puts both variances on the same scale:

$$\mathrm{Var}(\text{linear part}) \sim \frac{4c}{p+1}[\log(\theta^{-2})]^{2p+1},$$

$$\mathrm{Var}(\text{quadratic part}) \sim 2[\log(\theta^{-2})]^{2p}.$$

The linear variance is larger by one logarithmic factor, with power $2p+1$ rather than $2p$, so it dominates. Therefore,

$$\mathrm{Var}(\Xi_{n,w}) \sim \frac{4c}{p+1}[\log(\theta^{-2})]^{2p+1}.$$

Proof. See Appendix C.2.

5 Main concentration theorem

Theorem 5.1 (Concentration away from the crossing boundary)**.** For fixed $p > 0$ and $c \in (0, \infty) \setminus \{1\}$,

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \sim \frac{4c(p+1)}{(1-c)^2 \log(\theta^{-2})} \to 0.$$

Thus weighted RV selects Model 1 with probability tending to one for $c < 1$, and Model 0 with probability tending to one for $c > 1$. At $c = 1$, the leading mean vanishes.

Proof. See Appendix C.3.

Appendix

A Population risk and weighted-RV formulas

A.1 Population risk

Let $Y_{\text{new}} = \theta + \xi_{\text{new}}$ be independent of the training sample, with $\xi_{\text{new}} \sim N(0,1)$. Then $Y_{\text{new}} - \bar{Y}_n = \xi_{\text{new}} - n^{-1}\sum_{j=1}^n \xi_j$ has mean zero and variance $1 + 1/n$, whereas $\mathbb{E}Y_{\text{new}}^2 = 1 + \theta^2$. Subtracting the risks gives $\Delta_R(n) = 1/n - \theta^2$, negative exactly when $n > \theta^{-2}$. The first such integer is $\lfloor \theta^{-2} \rfloor + 1 = \theta^{-2} + O(1)$; equality of risks is a tie.

A.2 Weighted expectation, monotonicity, and crossing

Independence of Y_i and its training sample gives

$$\mathbb{E}\{(Y_i - \bar{Y}_{i-1})^2 - Y_i^2\} = \frac{1}{i-1} - \theta^2.$$

Linearity of expectation proves the stated weighted mean, without requiring independent validation losses. Dividing by positive total weight gives $r_n - \theta^2$. Each reciprocal is at least $1/n$, so $r_n \geq 1/n$ and a strict weighted crossing cannot precede n_{pop}. Moreover,

$$r_{n+1} = \frac{(\sum_{i=2}^{n+1} w_i) r_n + w_{n+2}/(n+1)}{\sum_{i=2}^{n+1} w_i + w_{n+2}} < r_n,$$

because $r_n \geq 1/n > 1/(n+1)$. Since $r_1 = 1 > \theta^2$, if $r_n \to 0$ there is a first strict crossing, and the inequality $r_n < \theta^2$ persists thereafter.

B Log-power weight sums and crossing scale

B.1 Weight-sum asymptotics

All limits in this subsection take $n \to \infty$ with fixed $p > 0$. For fixed $0 < \delta < 1$, weights at $\delta n \leq i \leq n+1$ equal $(\log n)^p(1 + o(1))$ uniformly. Bounding the full sum above by $n[\log(n+2)]^p$ and below by this last fraction gives

$$1 - \delta \leq \liminf_{n\to\infty} \frac{\sum_{i=2}^{n+1} w_i}{n(\log n)^p} \leq \limsup_{n\to\infty} \frac{\sum_{i=2}^{n+1} w_i}{n(\log n)^p} \leq 1.$$

Let $\delta \downarrow 0$. Replacing p by $2p$ proves

$$\sum_{i=2}^{n+1} w_i^2 \sim n(\log n)^{2p}.$$

With $m = i - 1$, the reciprocal-weight summand is asymptotic to $(\log m)^p/m$. This comparison sum diverges and its summand is eventually decreasing. Thus

$$\sum_{i=2}^{n+1} \frac{w_i}{i-1} \sim \sum_{m=2}^{n} \frac{(\log m)^p}{m} \sim \int_2^n \frac{(\log x)^p}{x}\, dx \sim \frac{(\log n)^{p+1}}{p+1}.$$

Dividing the sums gives the formula for r_n, in particular $r_n \to 0$.

B.2 First strict crossing

Fix $0 < \varepsilon < 1$ and set

$$n_- = \left\lfloor \frac{(1-\varepsilon)\theta^{-2}\log(\theta^{-2})}{p+1} \right\rfloor, \qquad n_+ = \left\lceil \frac{(1+\varepsilon)\theta^{-2}\log(\theta^{-2})}{p+1} \right\rceil.$$

Then

$$\log n_\pm = \log(\theta^{-2}) + \log\log(\theta^{-2}) - \log(p+1) + \log(1 \pm \varepsilon) + o(1),$$

so $\log n_\pm / \log(\theta^{-2}) \to 1$ and

$$\frac{r_{n_\pm}}{\theta^2} \sim \frac{\log n_\pm}{(p+1)n_\pm\theta^2} \longrightarrow \frac{1}{1 \pm \varepsilon}.$$

Since $r_n \to 0$, Appendix A.2 ensures existence for every admissible θ. For sufficiently small $|\theta|$, the limits above and monotonicity give $n_- < n_{\mathrm{RV},w} \le n_+$. Rounding changes each endpoint by at most one. Divide by $\theta^{-2}\log(\theta^{-2})/(p+1)$ and let $\varepsilon \downarrow 0$ to prove $n_{\mathrm{RV},w} \sim \theta^{-2}\log(\theta^{-2})/(p+1)$. The identity $\log(\theta^{-2}) = 2\log(|\theta|^{-1})$ and $n_{\mathrm{pop}} \sim \theta^{-2}$ give the crossing formulas.

C Mean, variance, and concentration on the log-weight scale

C.1 Mean

Appendix B.1 supplies deterministic remainders $o(\log n)$ and $o(n)$, depending only on n and fixed p, such that

$$\mathbb{E}\Xi_{n,w} = (\log n)^p \left\{ \frac{\log n}{p+1} + o(\log n) - \theta^2[n + o(n)] \right\}.$$

This proves the general expansion without a restriction on $n\theta^2$. Under (1), $n \sim c\theta^{-2}\log(\theta^{-2})/(p+1) \to \infty$, and

$$n\theta^2 \sim \frac{c}{p+1}\log(\theta^{-2}), \qquad \log n = \log(\theta^{-2}) + \log\log(\theta^{-2}) + \log\frac{c}{p+1} + o(1).$$

In particular $\log n \sim \log(\theta^{-2})$ and $n\theta^2 = O(\log n)$, so $\theta^2 o(n) = o(\log n)$. Hence

$$\mathbb{E}\Xi_{n,w} = (\log n)^p \left\{ \frac{\log n}{p+1} - n\theta^2 \right\} + o\big((\log n)^{p+1}\big).$$

This additive expansion holds also at $c = 1$. For $c \neq 1$, its leading coefficient is nonzero, giving the mean equivalent in Section 4.2. At $c = 1$ it proves only $\mathbb{E}\Xi_{n,w} = o([\log(\theta^{-2})]^{p+1})$.

C.2 Variance

In this proof write $S_m = \sum_{j=1}^m \xi_j$ and $f(x) = [\log(x+2)]^p$, so $w_{m+1} = f(m)$. Expanding one loss gives

$$(Y_{m+1} - \bar{Y}_m)^2 - Y_{m+1}^2 = \frac{S_m^2}{m^2} - \frac{2\xi_{m+1}S_m}{m} - 2\theta\xi_{m+1} - \theta^2.$$

Let $Q_{n,w}$ denote the centered quadratic Gaussian part of the score. The centered score is $Q_{n,w}$ plus $-2\theta\sum_{i=2}^{n+1} w_i\xi_i$, where

$$Q_{n,w} = \sum_{m=1}^{n} f(m)\left\{ \frac{S_m^2 - m}{m^2} - \frac{2\xi_{m+1}S_m}{m} \right\}.$$

Each centered quadratic monomial has zero covariance with each ξ_i: Gaussian third moments vanish and independent centered factors have zero expectation. Thus the quadratic and linear terms are uncorrelated. Collect the centered-square and pair-product coefficients as

$$a_t = \sum_{m=t}^{n} \frac{f(m)}{m^2} \quad (1 \le t \le n), \qquad b_l = \sum_{m=l}^{n} \frac{f(m)}{m^2} - \frac{f(l-1)}{l-1} \quad (2 \le l \le n+1),$$

with an empty sum equal to zero. Then

$$Q_{n,w} = \sum_{t=1}^{n} a_t(\xi_t^2 - 1) + 2\sum_{l=2}^{n+1} b_l \sum_{t=1}^{l-1} \xi_t\xi_l.$$

Distinct monomials in this expression are uncorrelated; centered squares have variance 2 and pair products have variance 1. Consequently,

$$\mathrm{Var}(\Xi_{n,w}) = 2\sum_{t=1}^{n} a_t^2 + 4\sum_{l=2}^{n+1}(l-1)b_l^2 + 4\theta^2\sum_{i=2}^{n+1} w_i^2. \tag{2}$$

We now estimate the quadratic part as $n \to \infty$ with fixed $p > 0$. Integral comparison gives $a_t \le Cf(t)/t$ for a constant C depending only on p, hence $\sum_t a_t^2 = O(1)$ since $\sum_t f(t)^2/t^2 < \infty$. Define the tail and residual coefficients

$$T_n = \sum_{m=n+1}^{\infty} \frac{f(m)}{m^2}, \qquad B_l = \sum_{m=l}^{\infty} \frac{f(m)}{m^2} - \frac{f(l-1)}{l-1}.$$

Integral comparison and integration by parts yield

$$T_n \sim \frac{f(n)}{n}, \qquad |B_l| \le C\left\{\frac{(\log l)^{p-1}}{l} + \frac{(\log l)^p}{l^2}\right\} \quad (l \ge 3).$$

Indeed, the sum–integral error is $O(f(l)/l^2)$, and

$$\int_l^\infty \frac{f(x)}{x^2}\,dx = \frac{f(l)}{l} + \int_l^\infty \frac{f'(x)}{x}\,dx, \qquad \int_l^\infty \frac{f'(x)}{x}\,dx = O\!\left(\frac{(\log l)^{p-1}}{l}\right).$$

For completeness, $f'(x)/x = O((\log x)^{p-1}/x^2)$, and for any fixed real q,

$$\int_l^\infty \frac{(\log x)^q}{x^2}\,dx = \frac{(\log l)^q}{l}\int_0^\infty e^{-v}\left(1 + \frac{v}{\log l}\right)^q dv \sim \frac{(\log l)^q}{l}.$$

Dominated convergence applies: for $q \le 0$ the factor is at most 1, and for $q > 0$ it is bounded by $(1+v)^q$ when $\log l \ge 1$. Taking $q = p-1$ proves the derivative bound. It also gives $\int_l^\infty f(x)/x^2\,dx = f(l)/l\{1 + O(1/\log l)\}$, which proves the tail and a_t estimates. The sum–integral bound applies for large l because $f(x)/x^2$ is eventually decreasing. The shift from $f(l)/l$ to $f(l-1)/(l-1)$ is $O(f(l)/l^2)$. Thus, writing $D_n = \sum_{l=2}^n (l-1)B_l^2$,

$$D_n = \begin{cases} O(1), & 0 < p < 1/2, \\ O(\log\log n), & p = 1/2, \\ O((\log n)^{2p-1}), & p > 1/2, \end{cases} \qquad D_n = o(f(n)^2).$$

These bounds follow by integrating $(\log x)^{2p-2}/x$. For $l \le n$, $b_l = B_l - T_n$, whereas $b_{n+1} = -f(n)/n$. Cauchy–Schwarz now gives

$$\begin{aligned}\sum_{l=2}^{n}(l-1)b_l^2 &= T_n^2\frac{n(n-1)}{2} + O\Big(D_n + T_n n\sqrt{D_n}\Big) \\ &\sim \frac{f(n)^2}{2}.\end{aligned}$$

The endpoint contributes $nb_{n+1}^2 = f(n)^2/n = o(f(n)^2)$. Since $f(n) \to \infty$, the bounded a_t contribution is also negligible. Therefore

$$\mathrm{Var}(Q_{n,w}) \sim 2f(n)^2 \sim 2(\log n)^{2p}.$$

Finally, Appendix B.1 gives $\sum_i w_i^2 \sim n(\log n)^{2p}$. On scale (1), $n\theta^2 \to \infty$, so the linear term in (2) dominates the quadratic term. Substituting the limits from Appendix C.1 proves the full variance equivalent in Section 4, also for $c = 1$.

C.3 Ratio and selection

For $c \neq 1$, the mean is eventually nonzero. Dividing the variance equivalent by the squared mean equivalent gives

$$\frac{\mathrm{Var}(\Xi_{n,w})}{\{\mathbb{E}\Xi_{n,w}\}^2} \sim \frac{\frac{4c}{p+1}[\log(\theta^{-2})]^{2p+1}}{\frac{(1-c)^2}{(p+1)^2}[\log(\theta^{-2})]^{2p+2}} = \frac{4c(p+1)}{(1-c)^2\log(\theta^{-2})} \to 0.$$

Writing $\mu = \mathbb{E}\Xi_{n,w} \neq 0$ in this proof, a wrong sign or tie implies $|\Xi_{n,w} - \mu| \geq |\mu|$. Therefore

$$\begin{cases} \Pr(\Xi_{n,w} \leq 0), & \mu > 0, \\ \Pr(\Xi_{n,w} \geq 0), & \mu < 0 \end{cases} \leq \Pr\{|\Xi_{n,w} - \mu| \geq |\mu|\} \leq \frac{\mathrm{Var}(\Xi_{n,w})}{\mu^2} \to 0.$$

The sign of the mean from Appendix C.1 then shows that weighted RV selects Model 1 with probability tending to one for $c < 1$, and Model 0 for $c > 1$. At $c = 1$, only $\mathbb{E}\Xi_{n,w} = o([\log(\theta^{-2})]^{p+1})$ follows; no ratio limit is asserted there.